



YAQZA AI

Predictive Maintenance System for Turbofan Engines

DEPI 2026 Graduation Project

The Industrial Problem

Traditional Maintenance

Fixed service intervals lead to wasted resources.
Reactive approaches result in catastrophic costs when unexpected failures occur mid-operation.

Parts: \$150,000 - \$400,000

Downtime: \$75,000 - \$200,000

**Cost per emergency: up to
\$1M**

The YAQZA Solution

A proactive, data-driven approach that predicts failures before they happen, optimizing scheduling and dramatically cutting operational costs.

Early Warning: 60+ Flight Cycles

Cost Reduction: 90% Savings

Safety: Guaranteed passenger security

NASA CMAPSS Dataset



FD001 & FD003

Sea level operating conditions.
Covers High Pressure Compressor (HPC) and Fan degradation. The gold standard foundation for prognostics research.



FD002 & FD004

Highly complex simulations across 6 mixed operating conditions (altitude, speed, power).
Considered the most challenging predictive maintenance datasets.



21 Sensors

Each row represents one operating cycle, capturing 21 critical sensor measurements and 3 operational settings, tracking the engine continuously until failure.

Engine Architecture

A modern turbofan engine consists of 5 main sections: the Fan (LPC), High Pressure Compressor (HPC), Combustor, High Pressure Turbine (HPT), and Low Pressure Turbine (LPT).

N1 Shaft (Low Pressure): Connects Fan to LPC and LPT, spinning at 3,000 to 5,000 RPM.

N2 Shaft (High Pressure): Connects HPC to HPT, spinning at an intense 10,000 to 15,000 RPM.

Key Sensor Monitoring:

- LPC & HPC Outlet Temperatures
- HPC Outlet Pressure
- Physical Core Speeds (N1/N2)
- Fuel Flow Ratio



Degradation Timeline



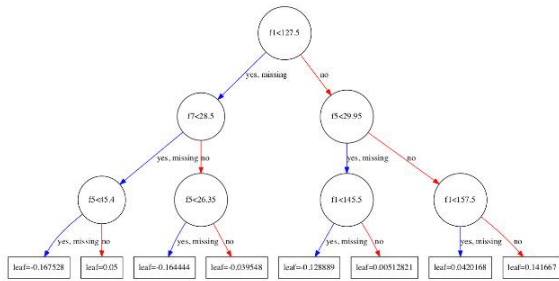
***The Compensation Trap:** As the HPC degrades, the engine burns more fuel to maintain thrust. This hides the real mechanical problem from simple fuel-flow monitors, requiring multi-sensor predictive models.*

8-Step ML Pipeline

-  **Scale per Condition & Baseline Removal:** KMeans clustering identifies 6 operating conditions for accurate scaling. Subtracted baselines isolate true degradation from individual engine variance.
-  **Low Variance Removal & MinMax Scaling:** Static sensors carry zero predictive value. Features are reduced from 24 to 14, then scaled proportionally to ensure unified model weighting.
-  **Rolling Window & TSFresh Extraction:** A 30-cycle sliding window captures critical temporal patterns. Over 100 features (statistical, trend, frequency, temporal) are extracted per sequence.
-  **FDR Feature Selection & PCA:** Hypothesis testing isolates statistically significant features ($p < 0.05$). Principal Component Analysis compresses these into 10-20 components containing 95% of the variance.

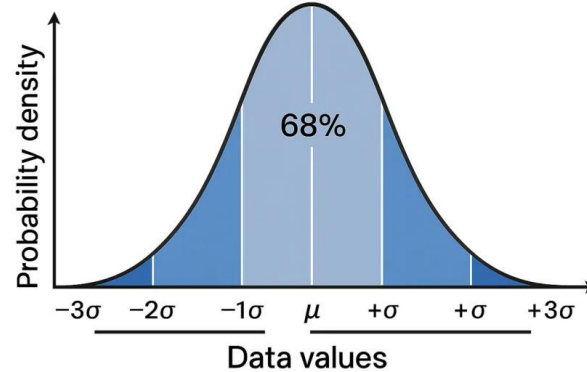
$$PC_1 = w_1 x_1 + w_2 x_2 + \dots + w_n x_n$$

5-Model Ensemble



XGBoost (Primary)

Our top performer. Utilizes sequential error correction and gradient boosting. Extremely fast inference makes it ideal for real-time production.



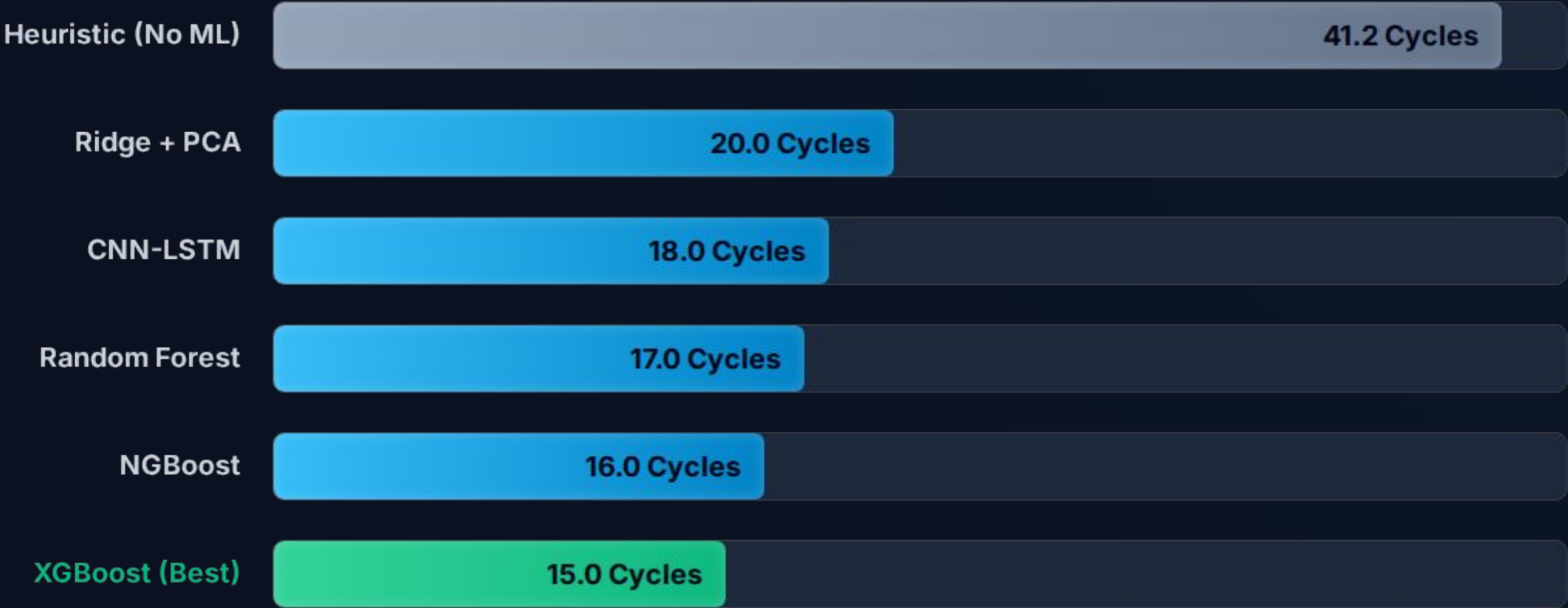
NGBoost

Provides critical probabilistic outputs. Instead of a single number, it yields a RUL distribution and confidence intervals for risk-aware scheduling.

CNN-LSTM

Our deep learning module. Learns end-to-end directly from raw sequences, capturing complex spatiotemporal patterns without manual feature engineering.

Model Performance (RMSE)



Note: Lower RMSE indicates superior accuracy. XGBoost achieves a prediction error of just 15 cycles—2.5 times more accurate than traditional heuristic baselines.

API Architecture

Production-Ready Ecosystem

The client layer interfaces seamlessly with our robust FastAPI Gateway, managing dynamic routing, authentication, and aggressive rate limiting.

The high-availability Data Layer combines SQLite at the edge, Redis for microsecond caching, and S3 for persistent backups.

Processed features instantly feed our Ensemble Consensus layer, triggering the Gemini AI module to auto-generate bilingual (En/Ar) diagnostic reports in real-time.



API Endpoints

Endpoint	Method	Primary Function
/ingest	POST	Real-time IoT/SCADA sensor data ingestion & processing.
/predict/{id}	GET	Rapid ensemble consensus Remaining Useful Life (RUL) prediction.
/predict/{id}/model/{name}	GET	Fetch isolated predictions and feature importance from a specific model.
/compare	GET	Identifies model disagreements and variance indicating high uncertainty.
/analyze	GET	Generates comprehensive bilingual (English/Arabic) AI maintenance reports.

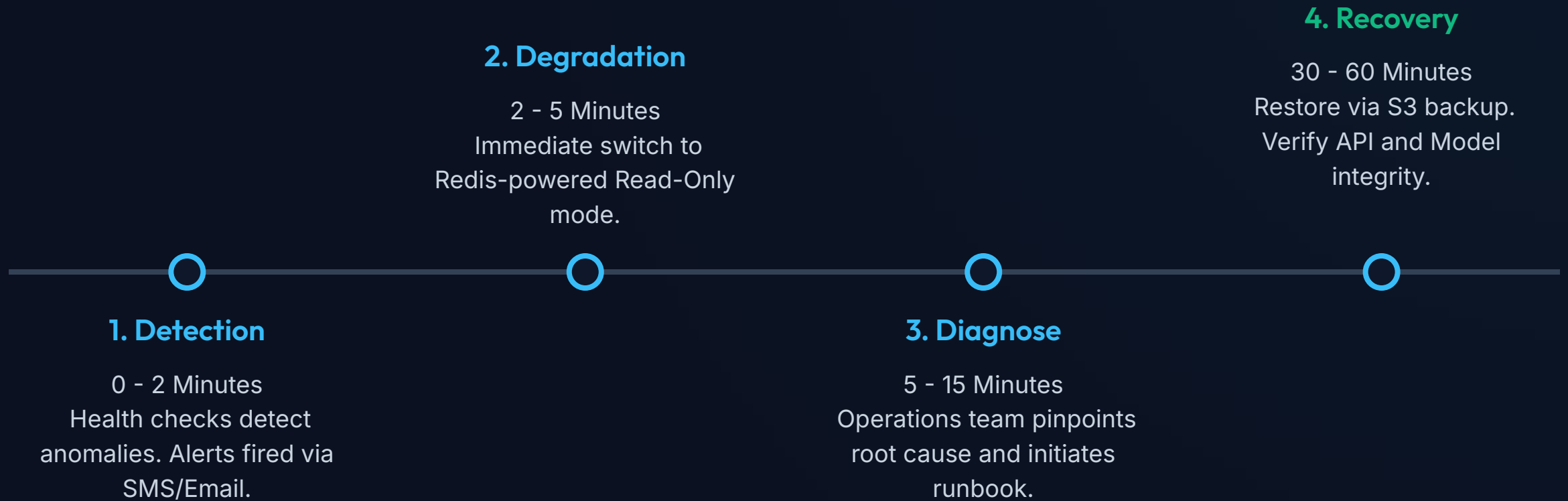
Production Pipeline

-  **Development & CI/CD:** Rapid iteration using Python/FastAPI natively. GitHub Actions automates aggressive testing suites, code linting, and seamless Docker image deployment to the registry.
-  **Staging Environment:** Full containerized deployment rigorously tested against CMAPSS test data. High-stress load testing via k6 guarantees performance under massive IoT data influx.
-  **Production Kubernetes:** Deployed on EKS/GKE with 3 core replicas for zero-downtime high availability. Horizontal Pod Autoscalers (HPA) instantly adapt to variable airline fleet loads.
-  **Optimized ML Serving:** All 5 predictive models reside in active memory. Leveraging ONNX runtime guarantees sub-100 millisecond inference, enabling true real-time threat detection.

Resilient Architecture

4 Graceful Degradation Levels • RTO < 1h • RPO < 6h • Failover < 5m

DB Failure Response



Cost Analysis

Without Yaqza (Status Quo)

Engines run blindly until failure. Unscheduled removals wreak havoc on logistics, heavily impacting the bottom line and compromising passenger safety.

Emergency Replacement: \$500,000

Unscheduled Removal: \$150,000

Fines & Insurance: \$100,000

Total Cost: \$750,000 / Incident

With Yaqza Integration

Failures are flagged 60+ cycles ahead. Supply chains execute just-in-time logistics. Downtime is fully planned, eliminating emergencies.

Scheduled Maintenance: \$50,000

Yaqza Software Cost: \$120,000 (5 Yrs)

Net Savings: \$580,000

ROI: 3.4x

Competitor Matrix

Feature Capability	Yaqza AI	OEMs (GE / RR)	Legacy Solutions
5-Model Machine Learning Ensemble	✓ Exclusive	✗	✗
Bilingual AI Advisor (Eng/Ar)	✓ Exclusive	✗	✗
Proprietary OEM Engine Lock-In	✓ NONE (Open)	✗ YES	Varies
System Deployment Timeline	✓ Hours	🕒 Months	🕒 Weeks
Starting Price Point (Monthly)	\$ 500+	\$ 10,000+	\$ 3,000+

Return on Investment

3.75x

ROI Multiplier

Massive Value Creation

Every \$1 invested in Yaqza yields \$3.75 in saved emergency costs and prevented downtime.

For a mid-sized fleet of 50 turbofan engines, utilizing our open-architecture consensus model equates to annual savings exceeding **\$2 Million USD**.

SaaS Business Model



Starter

\$500 / Month

Targeted at small MROs. Monitors up to 5 engines. Delivers basic RUL predictions, standard English reports, and essential email alerting capabilities.



Professional

\$1,500 / Month

The core tier for mid-size fleets. Up to 50 engines. Unlocks the full 5-model consensus, API/SCADA integration, and the Gemini Bilingual AI Advisor.



Enterprise

\$5,000+ / Month

Unlimited monitoring for major global airlines. Offers advanced digital twin integration, custom ML training loops, and rigorous 99.9% uptime SLAs.

Meet Our Team



Aya Yasser



Basmala Saeed



Esraa Mohamed



Ahmed Badawy



Omar Abdullah

Thank You

We appreciate your time and attention.



OPEN SOURCE PROJECT

github.com/basmalaeltabakh/Yaqza